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JAISP: A Joint Rubin–Euclid Foundation Model for Cross-Instrument Source Detection and Astrometry

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ABSTRACT

Combining Rubin and *Euclid* imaging at the pixel level promises systematics control beyond the reach of either survey on its own, but requires jointly treating two instruments, and ten wavebands, that differ in resolution, sampling, depth, and noise. We introduce Jaisp (Joint AI Survey Processing), a ∼9-million-parameter foundation model that learns a shared representation of the ten wavebands (Rubin *ugrizy* and *Euclid* VIS, Y, J, H) by self-supervised masked-band reconstruction: each band is predicted from the other nine, with each instrument encoded at its delivered sampling. Trained on overlapping public imaging of a single deep field, the encoder is then frozen, and every downstream measurement is a small head reading its features. In detection, a head trained on lightweight classical labels matches the published VIS-detected *Euclid* catalog on held-out sky, at 94% completeness and 94% purity, and the ten-band information pushes detections ≈0.3 VIS mag deeper than the single band supports. In astrometry, a position head carries VIS-quality centroiding into the other bands: cross-survey scatter falls from ∼50 to 7–11 mas, and injected sources are recovered to 19 mas at S/N = 5, near the floor set by the VIS labels. Beneath the per-source scatter lies the concordance field: a coherent 9–10 mas pattern between the two independently Gaia-anchored astrometric solutions, mostly a single shift common to every band. Applied to a second deep field with no retraining, the entire stack reproduces this performance. Photometry, shape, and redshift heads, in preparation, will read the same frozen features.

Keywords: Astrometry — Surveys — neural networks — Astronomy data analysis

1. INTRODUCTION

Cosmology has entered a regime where its hardest questions are limited by systematics rather than by statistics. The current generation of wide-field surveys is measuring the growth of structure and the expansion history precisely enough that the residual tensions between probes can no longer be dismissed as noise, yet cannot be confidently attributed to new physics rather than to unmodeled measurement error (**?**). Breaking this impasse depends on the next generation of surveys not merely collecting more photons, but controlling systematics far more tightly than before. The Vera C. Rubin Observatory's Legacy Survey of Space and Time (LSST; **?**), ESA's *Euclid* mission (**?**), and the *Nancy Grace Roman Space Telescope* (**?**) will observe overlapping sky from ∼0.3 to 2 μm at sub-arcsecond resolution. Individually powerful, the observatories in combination can calibrate and cross-check one another, suppressing survey-specific systematics that no single survey can control alone. The case for combining them at the pixel level rather than the catalog level was laid out by the Joint Survey Processing study (**?**): at these depths up to a quarter of the sources in first-year ground-based stacks, and up to half at full depth, are blended at seeing-limited resolution, and astrometric scatter between surveys propagates directly into the error of any joint photometry, which is why that study placed a common astrometric and photometric reference across the surveys, its concordance frame, first among its requirements. Exploiting the overlap at the pixel level, however, means confronting heterogeneous PSFs, samplings, and noise properties simultaneously, a regime where classical pipelines strain.

Machine and deep learning have moved quickly from peripheral tools to central ones in survey astronomy (e.g., **???**), and there is a growing case that much of the next decade of pixel-level processing will rely on learned components alongside explicitly modeled ones.

This is not, however, a shortcut. The classical pipelines these surveys rely on represent decades of accumulated, problem-specific effort: careful PSF models, distortion solutions, deblenders, and uncertainty propagation, each refined against the failure modes of real data. Reaching, and then surpassing, that level of fidelity with learned methods requires comparable investment, because the understanding encoded in the classical solution is precisely what tells us how to design, constrain, and validate its learned replacement. Once that effort is invested, the payoff is a more data-driven pipeline: inference is fast, and the representation is shaped by the pixels themselves rather than by parametric assumptions, absorbing structure such as cross-instrument PSF differences, wavelength-dependent morphology, and correlated noise that classical stages can only approximate.

This does not mean every problem calls for a single, large foundation model. Much recent work comes instead from compact, purpose-built networks designed around a specific measurement and its systematics: generative and physics-informed models for deblending, super-resolution, and the realistic image simulations needed for shear calibration (e.g., **????**). In these, the architecture follows directly from the problem. Foundation models occupy a complementary niche: rather than solving one task, they learn a general representation once through self-supervised pretraining and amortize it across many downstream tasks. This paradigm has begun to take hold in astronomy, through cross-modal and autoregressive models trained on large galaxy samples and the multi-survey datasets assembled to support them (e.g., **???**). Building on this direction, we apply it to a problem central to joint survey processing: combining two imaging instruments at the pixel level, across mismatched resolutions and overlapping but non-identical bandpasses.

In this work we introduce JAISP (Joint AI Survey Processing), which learns a single shared representation of Rubin and *Euclid* imaging, using only the first public data release of each, through self-supervised pretraining, and reuses it across several downstream measurements. We work from already-processed survey mosaics rather than individual exposures, which lets us concentrate on the joint representation itself; the same approach could in principle be pushed earlier in the pipeline, toward less-processed data where more of the instrumental signal survives. A foundation model is trained once to capture the cross-instrument spatial and spectral structure that joint processing requires; its frozen features then drive lightweight task-specific heads. We focus here on source detection and astrometry, benchmarked against classical baselines; the same frozen features can serve many further downstream measurements, including photometry, shape measurement, and redshift estimation, which we develop in forthcoming work. Even at the mosaic level, we find that the learned representation transfers *Euclid*'s sharp localization to Rubin's bands and to faint sources, improves source detection by combining all ten bands, and measures the relative astrometric offsets between two independently *Gaia*-aligned surveys. These results hold on a separate deep field with no retraining, a first test of the representation's transferability across the sky.

This paper is organized as follows. Section 2 describes the Rubin and *Euclid* imaging; Section 3 presents the foundation model and the frozen-feature head design, and validates the learned representation; Sections 4 and 5 present the detection and astrometry heads; Section 6 tests transfer to a separate deep field; and Section 7 summarizes the results and discusses implications and next steps.

2. DATA

The dataset for this work combines overlapping imaging from Rubin and *Euclid* over two widely separated Euclid Deep Fields. Training uses the Euclid Deep Field Fornax (EDF-F), which coincides with the Extended Chandra Deep Field South (ECDFS) in the Rubin footprint; the Euclid Deep Field South (EDF-S) is reserved entirely for the out-of-distribution transfer test of Section 6 and is never seen during training. For each field we assemble a common ten-band stack: the six Rubin optical bands *ugrizy* together with *Euclid* VIS and the three NISP near-infrared bands Y, J, and H (Figure 1). All imaging is drawn from currently public data releases. The region used here is a small part of the available Rubin–*Euclid* overlap: it is a single Rubin tract within one deep field, while the deep overlap spans several tracts per field and multiple fields, leaving substantial room to scale the same approach to larger areas as deeper releases arrive.

Rubin imaging is drawn from the Legacy Survey of Space and Time Data Preview 1 (DP1; **?**), the first release from the Rubin Observatory. DP1 comprises coadded *ugrizy* imaging obtained with the Rubin Commissioning Camera (LSSTComCam) during on-sky commissioning in late 2024, covering seven non-contiguous fields. Of these fields, ECDFS is the deepest and most uniformly observed, and we use its coadds in all six bands. Within ECDFS we use a single Rubin tract (tract 5063), spanning six patches, which covers the region of common Rubin–*Euclid* depth used here. The coadds are sampled at $0.2'' \,\mathrm{px}^{-1}$; the median point-spread function is broad by space-based standards, with

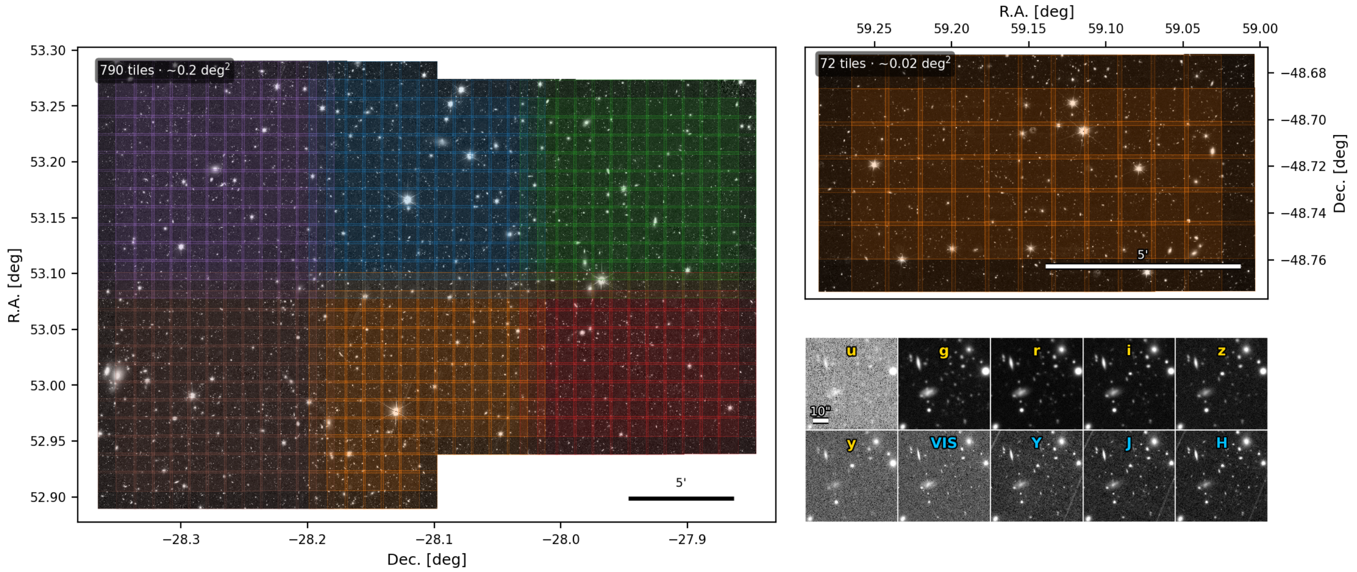

Figure 1. Data and survey footprints. *Left:* the ECDFS field used for training, where Rubin and *Euclid* coverage overlap: a single Rubin tract (5063) spanning six patches, tiled into 790 512×512 cutouts (stride 256 px, 50% overlap) over $\sim$0.2 deg^2, shown on the *Euclid* VIS mosaic and tinted by patch. *Top right:* the EDF-S field, a widely separated region held out to test out-of-distribution transfer with no retraining: 72 tiles ($\sim$0.02 deg^2). *Bottom right:* the ten input bands for one matched tile, Rubin *ugrizy* (gold labels) and *Euclid* VIS and NISP $Y/J/H$ (blue labels); each panel is a $1'$ field. Rubin imaging is sampled at $0.2''$ px^{-1} and the *Euclid* bands are drawn from the $0.1''$ px^{-1} MER mosaics, giving the joint ten-band stack used throughout this work.

a full width at half maximum of roughly $1''$ at this early commissioning stage. *Euclid* VIS provides the sharp spatial reference throughout this work. The 5σ point-source coadd depths in this field reach $\approx$26 in g and r, with $u \approx 24.6$ and $y \approx 23.1$ the shallowest; the full per-band values and processing details are given in the DP1 release paper.

Euclid imaging is drawn from the Quick Data Release 1 (Q1; **?**), the first public *Euclid* release, which covers the three Euclid Deep Fields, including EDF-F and EDF-S used here. Despite the deep-field designation, Q1 provides a single visit of each field, reaching approximately the depth of the Euclid Wide Survey; the full deep-field depth will accumulate over repeated visits during the mission. We use the visible channel VIS (**?**) and the three NISP near-infrared bands Y, J, and H (**?**), taken from the background-subtracted MER mosaics produced by the *Euclid* MERge processing function (**?**). All four bands are delivered on a common $0.1''$ px^{-1} grid, with VIS providing the highest resolution ($\sim 0.16''$) and the NISP bands resampled onto the same grid; the 5σ point-source depths are ≈ 26 in VIS and ≈ 24 in each NISP band. Per-pixel variance maps accompany every band and are used for noise normalization throughout (Section 3).

Using the MER mosaics means inheriting that preprocessing: background subtraction, and in particular the resampling of the native near-infrared NISP frames onto the finer VIS pixel grid, are applied before our pipeline sees the data. This is a deliberate, conservative starting point that keeps the input fully reproducible from public products. Q1 does already include the near-infrared mosaics at their native NISP sampling, but as science images alone, without the per-pixel variance maps our pipeline requires for noise normalization. Native-sampling products with their full supporting maps, expected in deeper releases beginning with DR1, would let a joint model ingest them directly and perform the cross-instrument registration itself, reducing the preprocessing carried over from the survey pipeline; we return to this in Section 7.

For each field we lay the imaging out on a common tile grid. Rubin tiles are 512×512 pixels ($\sim 102''$ on a side at $0.2''$ px^{-1}), with the matching *Euclid* region cut at twice the sampling; the ten bands of each tile are spatially aligned to a common frame. Tiles are placed on a 256-pixel stride, so adjacent tiles overlap by 50% and a given point on the sky falls in up to four tiles. The training field yields 790 matched Rubin–*Euclid* tiles over $\sim$ 0.2 deg^2; the held-out EDF-S field provides a further 72 matched tiles (~ 0.02 deg^2), used only for the transfer test of Section 6. This overlap is useful during pretraining, where the same source appears at different positions relative to the tile edge and so provides free augmentation, and during detection, where sources that sit near the edge of one tile fall near the center of a neighbor.

The overlap does, however, require care in evaluation. Because the tiles are cut from the same underlying coadd mosaics, the pixels shared by two overlapping tiles are identical rather than independent re-observations. A naive tile-level train/validation split would therefore place bit-identical copies of the same sky, carrying identical labels, on both sides of the split, making validation metrics artificially inflated. The in-field validation splits used for model selection and leakage checks are therefore spatially disjoint at the level of Rubin patches, so that no sky region and no source appears in both training and validation. Some diagnostic figures below pool the full ECDFS field to characterize the final frozen stack; where they do, we treat them as full-field measurements and compare them to patch-disjoint controls and to the fully held-out EDF-S transfer field.

3. ARCHITECTURE

The core of JAISP is a single foundation model that learns a shared representation of the Rubin and *Euclid* imaging, which is then reused by separate task-specific heads. The foundation model is trained by self-supervised masked-band reconstruction, a masked autoencoder (MAE; ?) applied across wavelength bands rather than spatial patches: from the multi-band input, one band is withheld and the model is required to predict its image from the remaining ones. No labels are used at this stage, since the supervision comes entirely from the requirement that the model reproduce held-out pixels, so the model can be trained on raw imaging alone. Once trained, the encoder is frozen and its features drive the downstream heads. This work develops the detection and astrometry heads, with photometry, shape-measurement, and redshift heads in preparation (Figure 2).

The architecture must reconcile a mismatch between the instruments. Rubin imaging is sampled at $0.2'' \,\mathrm{px}^{-1}$ across six optical bands ($ugrizy$), whereas *Euclid* VIS and the three NISP bands (Y, J, H) from MER are sampled twice as finely, at $0.1'' \,\mathrm{px}^{-1}$. Resampling both onto a single common grid would discard spatial information from the sharper instrument. The model instead keeps each instrument at its native resolution through separate encoder branches, and brings them together only once they reach a common physical scale.

Each band first passes through its own convolutional "stem," a small network that maps a single noisy band into a shared feature space while normalizing for that band's noise level. The per-band features then enter one of two parallel encoder branches, one for Rubin and one for *Euclid*, built from ConvNeXt blocks (??). ConvNeXt modernizes the classical convolutional network with design choices adapted from vision transformers (?), such as large depthwise kernels and inverted bottlenecks, matching transformer performance on vision benchmarks while remaining purely convolutional: it keeps the locality and translation equivariance natural to imaging data, and its cost scales linearly with image area rather than quadratically as self-attention does, which matters at the pixel counts of survey imaging. We therefore use convolutions wherever the sampling is fine and reserve explicit attention for the coarse fused grid below. Within each branch the feature maps are progressively reduced in spatial size and increased in depth, so that despite their different native sampling both branches arrive at a common $0.4'' \,\mathrm{px}^{-1}$ grid. The two streams are then summed with learned identity embeddings that let the network tell Rubin features from *Euclid* ones, and passed to a transformer bottleneck (?). Through self-attention, every location in the fused map can exchange information with every other, and it is here that cross-instrument and cross-band relationships are combined. Per-instrument decoders expand the bottleneck features back to native resolution to reconstruct the withheld band, with the target band specified to the decoder so that a single model can reconstruct any of the bands. The complete model contains roughly nine million parameters; layer-by-layer specifications are given in Figure 2. The $0.4'' \,\mathrm{px}^{-1}$ fused scale is a measured choice rather than a default. As a matched-token control, we also trained a foundation variant whose bottleneck grid is refined to $0.2'' \,\mathrm{px}^{-1}$ while the crop size is halved to keep the transformer token count fixed. Because the current encoder keeps at least one convolutional block in each native-resolution stream, this control is not an exact parameter-matched replica of the fiducial model; rather, it tests whether moving the fused tokens to $0.2'' \,\mathrm{px}^{-1}$ produces an obvious representation gain under comparable transformer cost. It does not: held-out band reconstruction and the decodability diagnostics of Section 3.3 are indistinguishable at the level relevant here, so we retain the coarser, cheaper fused scale.

Within each branch the per-band feature maps must be combined before the streams are fused. A natural default is to pool them into a single map, but doing so attenuates each band's contribution to the learning signal: averaging six Rubin bands, for example, divides each band's gradient by six, so the encoder is discouraged from forming sharp per-band features. The model instead concatenates the per-band features and projects them, preserving each band's identity through the en-

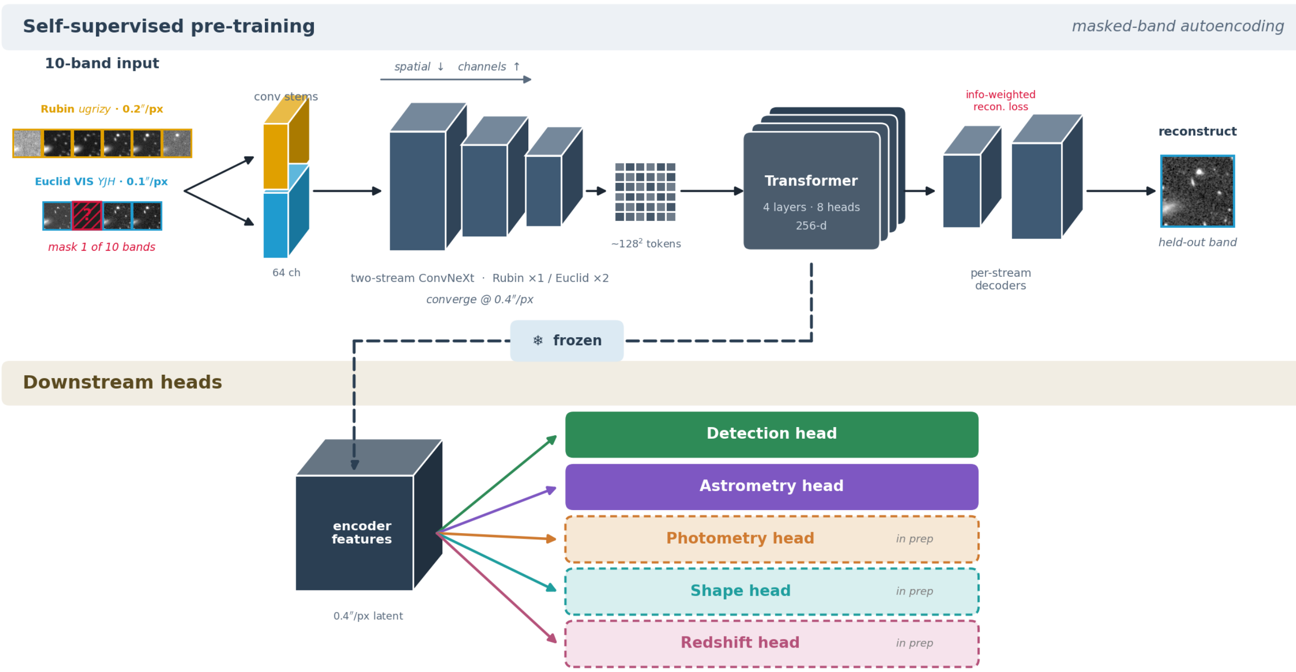

Figure 2. Model architecture. *Top:* self-supervised pretraining by masked-band autoencoding. The ten input bands, Rubin *ugrizy* at $0.2'' \mathrm{px}^{-1}$ and *Euclid* VIS YJH at $0.1'' \mathrm{px}^{-1}$, pass through per-band convolutional stems and a two-stream ConvNeXt encoder (Rubin and *Euclid* branches) whose feature maps shrink spatially and deepen in channels until they converge on a common $0.4'' \mathrm{px}^{-1}$ grid. A transformer bottleneck (4 layers, 8 heads, 256-d) mixes the $\sim 128^2$ fused tokens, each carrying joint Rubin–*Euclid* features for one $0.4'' \mathrm{px}^{-1}$ location, and per-stream decoders reconstruct the single held-out band under an information-weighted loss; the complete model totals ∼9.1M parameters. *Bottom:* for downstream tasks the encoder is frozen and its $0.4'' \mathrm{px}^{-1}$ latent features feed lightweight task heads, detection and astrometry in this work, with photometry, shape, and redshift heads in preparation. Input and reconstruction thumbnails show the same matched ECDFS tile as Figure 1.

coder at full strength. This keeps wavelength-dependent information, such as how a galaxy's morphology and effective PSF change from band to band, available to the representation, where the photometry and deblending heads later draw on it. When a band is withheld as the reconstruction target its input slot is zero-filled, and the projection learns to ignore the empty slot, which also lets the model operate when some bands are absent.

The reconstruction loss is computed in noise-normalized units and weighted toward pixels carrying source signal, so the model is judged on how well it reproduces astronomical structure rather than blank sky. Without this weighting the loss would be dominated by the empty background that fills most of every image, and the model could drive it down while reproducing sources poorly. A robust per-pixel loss is used, with additional weight on the bright cores of sources.

3.1. *From latent alignment to pixel reconstruction*

The masked-band objective was not the first design we tried, and the reconstruction approach was reached only after several alternatives failed to deliver the spatial precision the downstream tasks require. We summarize that history here because the negative results explain why reconstruction, rather than the more common alternatives, is the right objective for this problem.

The natural approach to relating two instruments is *representation alignment*: encode a Rubin image and the matching *Euclid* image each into a compact feature vector, and train the network so that the two vectors agree where they view the same sky. This is the principle behind contrastive learning (**?**) and Joint-Embedding Predictive Architectures (JEPA; **?**), both widely used in computer vision. We designed and tested several cross-instrument variants in this family: patch-level contrastive matching, an object-centric model that attempts to discover individual sources, and JEPA architectures operating on the full image at native resolution with increasingly strict constraints forcing features at corresponding positions to match.

These approaches shared a common limitation that is structural rather than a matter of tuning. Alignment objectives reward features that are *similar* but do not require the network to encode *where* sources are located: two image regions can map to nearly identical feature vectors while differing in the sub-pixel positions of the sources they contain, and sub-pixel position is the quantity that astrometry and detection depend on. A direct control made this explicit: a simple supervised convolutional network, trained directly on the cross-instrument alignment task, outperformed the best self-supervised JEPA features at substantially lower cost.

Pixel reconstruction removes this failure mode by construction. To predict a withheld band at the correct positions, brightness, and morphology, the representation has no choice but to preserve the spatial layout of every source, a constraint that similarity-based objectives never impose. This is why the masked-band objective, rather than alignment, underlies the model described above.

3.2. *Implementation*

The foundation model was trained on a single NVIDIA L40S GPU using `PyTorch` (?) with bfloat16 mixed-precision arithmetic. Optimization used AdamW (?) with gradient clipping at unit norm, a five-epoch linear warmup followed by cosine decay, and an effective batch size of four, for 150 epochs; the checkpoint with the lowest validation loss was retained.

Pre-training completed in roughly 8 wall-clock hours (∼8 GPU-hours). Once trained, the encoder is frozen and run once over the full dataset to extract and cache per-tile features; each downstream head is then a small network trained on these cached features rather than on the full imaging. The feature-extraction pass is the main per-dataset cost, but like the pre-training it is incurred only once and shared across all heads. Training was monitored with the Weights & Biases platform.

The reconstruction loss is evaluated in noise-normalized units, with each pixel weighted by a signal-and-gradient information map so that source pixels dominate over empty sky. The per-pixel term is a Charbonnier penalty (?), supplemented by an additional squared-error term on the highest-information "core" pixels of sources. The exact functional form and all coefficients are given in the public code. The complete implementation, including the foundation model, the reconstruction loss, the downstream heads, and the scripts that reproduce the analyses and figures in this paper, is publicly available.[3]

3.3. *Validation of the learned representation*

Before attaching any task-specific head, we verify that the frozen representation has captured the cross-instrument structure it was trained to learn. We do this in three ways: by checking that held-out bands can be

[3] https://github.com/xoubish/JAISP

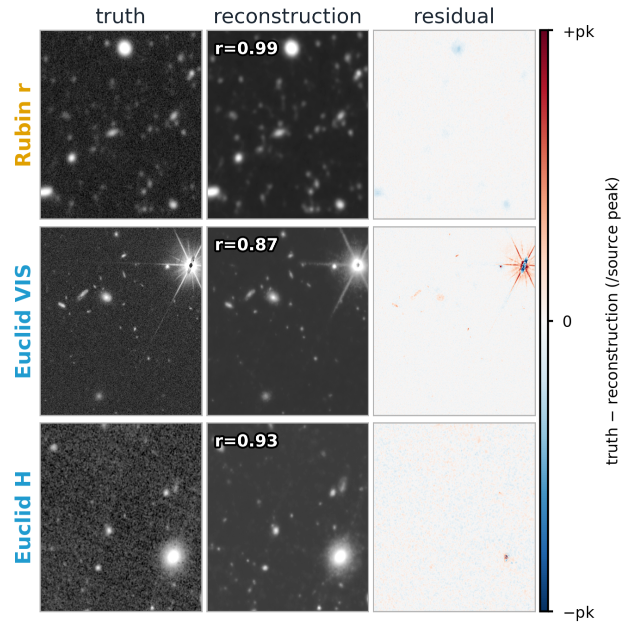

Figure 3. Masked-band reconstruction on three held-out ECDFS tiles. In each row a single band is withheld and reconstructed from the other nine: Rubin r (top), *Euclid* VIS (middle), and *Euclid* NISP H (bottom). Columns show the true band (left), the reconstruction (center), and the residual truth − reconstruction (right); panels are $48''$ on a side. The residual uses a diverging scale whose limits are the brightest source pixel of that row (±pk), so the near-white residuals indicate errors that are a small fraction of the signal. The annotated r is the Pearson correlation between truth and reconstruction over the brightest 10% of pixels; averaged across the 39 held-out tiles the mean per-band correlation is 0.99 (r), 0.77 (VIS), and 0.90 (H).

reconstructed faithfully, that the per-band information is recoverable from the frozen features a head would use, and that the model routes information between bands in a physically sensible way.

Reconstruction fidelity. Figure 3 shows masked-band reconstruction on validation tiles never seen during training. In each case a single band is withheld and predicted from the other nine, sampled across the full input range: a Rubin optical band, the high-resolution *Euclid* VIS channel, and a *Euclid* NISP near-infrared band are shown. The reconstructions reproduce the held-out sky faithfully, with residuals confined to the cores of the brightest point sources and the faint extended wings of large galaxies. Averaged over the held-out tiles, the truth-to-reconstruction correlation on the brightest tenth of pixels is 0.97–0.99 for the deep, spectrally interior Rubin bands ($griz$), and each departure from that level has a physical origin rather than an architectural one. The shallow bands u and y (0.64 and

0.84) are limited by a noise ceiling: their truth images are themselves noisy, which bounds the correlation any prediction can achieve, and u sits in addition at the blue edge of the ten-band range, where prediction becomes extrapolation. The NISP bands (0.85–0.90) draw chiefly on one another, as the attribution analysis below makes explicit. VIS, at 0.77, is the one channel sharper than everything used to predict it: reconstructing it is an implicit deconvolution, asked to supply spatial frequencies the other nine bands do not contain, and the bright-pixel metric concentrates on exactly the point-source cores where those frequencies matter most. Some of this fidelity is expected on general grounds: galaxy spectral energy distributions are smooth, and for bright, well-detected sources the flux in a missing band can be recovered to first order by linear interpolation between its neighboring filters (**?**). The reconstruction here carries more than that first-order transfer: it operates at every pixel rather than only on cataloged sources, it shapes its prediction to each source's morphology and surroundings instead of applying one global color law, and it crosses instruments, with their different pixel grids, samplings, and point-spread functions, so that a band from one telescope is predicted from the imaging of the other, something no within-instrument interpolation provides.

Decodability. Faithful reconstruction is a statement about the network as a whole: the held-out band is produced by the encoder's features and the trained decoder acting together, so its success shows only that the information exists somewhere along that path, possibly folded into the decoder's weights. A downstream head inherits none of the decoder; it consumes the frozen bottleneck features alone. The question that matters for everything that follows is therefore whether those features, by themselves, hold the per-band information in a form a simple head can read out. We test this with a linear probe (left panel of Figure 4): a ridge regression trained to predict each band's per-source aperture S/N from the frozen features at the source position. Nothing learned enters the setup beyond the frozen encoder itself: the sources are simply S/N > 5 peaks in the smoothed VIS image, and the quantity to be recovered is an aperture S/N measured in each band's native pixels at the peak position. The probe is evaluated under a spatial hold-out, so it is always tested on sources outside its training set, with features standardized per fold and the regression restricted per band to sources carrying signal in that band. All ten bands are recovered with high accuracy ($R^2 \approx$ 0.7–0.95), and a nonlinear probe adds little, showing the per-band photometric information is not only present but *linearly* available in the representation. We note that source brightness is an easy quantity to decode, so this is a check on representational richness rather than a strong claim about deep cross-instrument content; the more discriminating question is addressed next.

Information routing. The right panel of Figure 4 asks which bands the model actually draws on when reconstructing a given band. We measure this by input-gradient attribution, a standard interpretability technique whose idea is simple: the derivative of the reconstruction loss with respect to an input pixel tells us how much the prediction would change if that pixel were perturbed, so its magnitude records how strongly the model relies on that pixel. For each withheld target band we accumulate these sensitivities over the pixels of every remaining band, giving the fraction of the reconstruction that each source band drives. The dominant result is that, when *Euclid* VIS is available, it absorbs essentially all of the attribution: as the sharpest and deepest channel, it serves as a universal spatial scaffold on which every other band is reconstructed. Because this single-band dominance would otherwise mask all finer structure, the panel shown is a counterfactual with VIS removed, exposing how the remaining nine bands route information among themselves. To summarize each target band's reliance on others compactly, we report the effective number of contributing source bands, $n_{\mathrm{eff}} = 2^H$, where H is the Shannon entropy of that band's normalized attribution distribution; n_{eff} is a well-defined, assumption-free measure that equals one when a band is reconstructed from a single other band and grows as the reconstruction spreads its reliance across many. The Rubin optical bands have $n_{\mathrm{eff}} \approx$ 4–7, drawing on many spectral neighbors with the adjacent bands contributing most, whereas the three NISP near-infrared bands form a near-closed block ($n_{\mathrm{eff}} \approx 3$) that relies chiefly on one another and barely on Rubin. This mirrors the structure of real galaxy spectral energy distributions, where neighboring bands are most predictive of one another, and indicates that the representation has organized the bands by genuine physical relationships rather than arbitrary correlations. Together these checks establish that the frozen representation is faithful, that its per-band information is linearly accessible, and that it routes information between bands in a physically coherent way. It is therefore a sound basis for the task-specific heads developed in the following sections.

4. DETECTION

Source detection is the first measurement we build on the frozen foundation, and it asks a specific question: does the frozen latent carry multi-band source evidence beyond what a single sharp band already provides, and

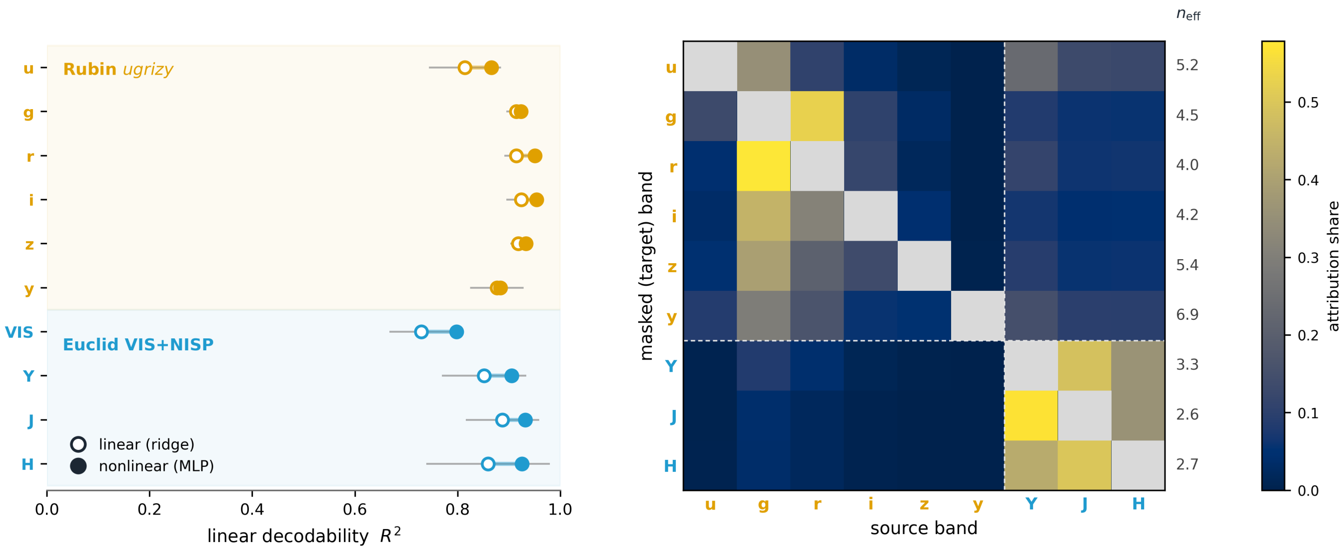

Figure 4. Two views of the frozen foundation representation. *Left (decodability):* a probe predicts each band's per-source aperture log S/N from the frozen bottleneck features at detected source positions (6-fold spatial hold-out, ∼770 sources; protocol in text). Open markers are a linear (ridge) probe, filled markers a nonlinear (MLP) probe, the connecting line is the nonlinear gain, and the gray bar is the fold scatter. All ten bands are linearly decodable ($R^2 = 0.7$–0.95); VIS reads back lowest only because sources are selected on VIS S/N, truncating its range. *Right (information routing):* input-gradient attribution, masking a target band (row) and backpropagating the reconstruction loss to each source band's pixels (row-normalized; the diagonal is self-attribution, shown gray). VIS is omitted because, when present, it absorbs ∼100% of the attribution; shown is the routing among the remaining nine bands. $n_{\rm eff} = 2^H$, with H the Shannon entropy of each row, is the effective number of source bands: Rubin bands draw on many sources ($n_{\rm eff} \approx 4$–7, spectral neighbors with a g/r hub, plus NISP), whereas the NISP bands form a near-closed block ($n_{\rm eff} \approx 3$, $Y/J/H$ mutually, with only ∼10% of their attribution reaching Rubin).

Table 1. 50% point-source injection depth d_{50} (VIS mag) for the two detection heads, measured with unresolved donors.

	CenterNet (fused)	StemCenterNet (stems)
d_{50}, all 10 bands	26.7	26.5
d_{50}, VIS only	26.4	26.5
multi-band gain	+0.3	+0.0

can a lightweight head built on it match classical detection and then surpass it? Detection on *Euclid* VIS alone is sharp and reliable, but it cannot use color or the Rubin and NISP channels in a learned way. We compare three source lists that differ only in what information they see: a classical VIS source-extraction list (SEP), which serves as the baseline and as the source of training labels, and two variants of a learned detection head built on the frozen foundation features. Comparing them isolates whether the representation has learned genuine multi-band source evidence, and whether that evidence is best expressed in the fused latent or in the higher-resolution per-band features.

Both variants share the same output design and differ only in the features they read. Following CenterNet (**?**), each predicts a dense source-evidence map over the tile, in which a true source is represented during training by a small Gaussian peak at its position rather than by a single hard pixel, so that nearby pixels receive a graded target suited to sub-pixel positions. A focal loss handles the extreme imbalance between the rare source pixels and the dominant empty sky. Alongside the evidence map each predicts a sub-pixel offset and a brightness proxy at every location. At inference the procedure is essentially learned peak-finding: local maxima above a confidence threshold are kept, and the offset is read off to refine each position. This is the same operation as classical peak-finding, but performed on a learned ten-band source-evidence image rather than on a single band.

The two variants differ in where they read that evidence. The fused-bottleneck variant (CenterNet) operates on the frozen bottleneck, where the foundation model has already mixed all ten bands, and so tests whether the representation has learned multi-band source evidence. The native-stem variant (StemCenterNet) operates instead on the per-band stem features at their original resolution, projected onto a common grid, and so tests whether finer spatial detail helps localization beyond what the coarser bottleneck retains. Each trains well under a million parameters on top of the frozen encoder (0.55M for the fused variant): the en-

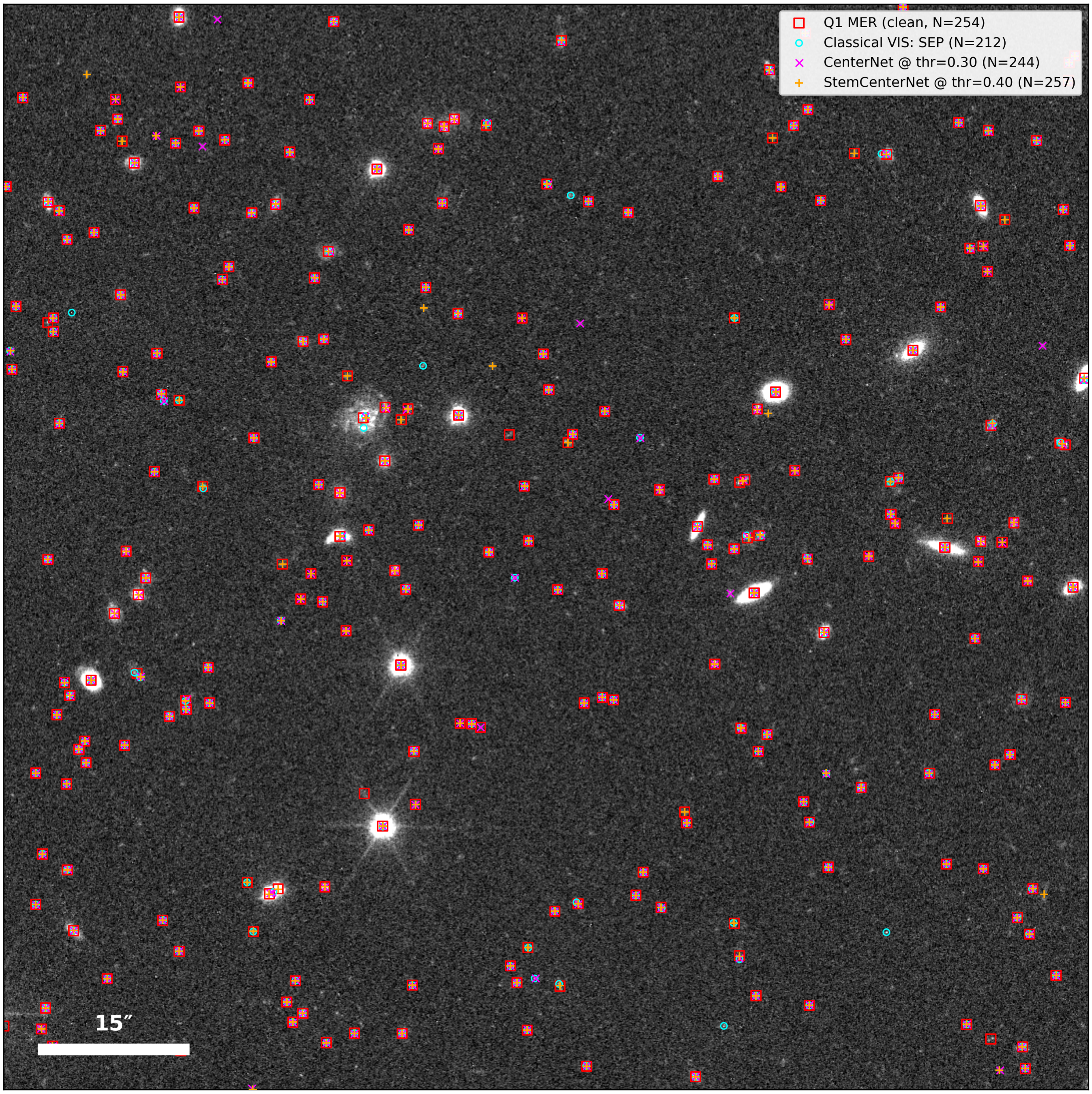

Figure 5. Detection on a single held-out *Euclid* VIS tile: the foundation head against classical and catalog references. A 1.7′ tile (`tile_x01792_y00000`, 0.1″ px^{-1}, from a spatially disjoint patch excluded from detection training), with four source lists overlaid. *Cyan circles:* the classical VIS source list (SEP extraction at 3σ, cleaned of diffraction-spike artifacts and duplicates). *Magenta crosses:* the foundation CenterNet head reading the frozen, fused ten-band bottleneck. *Orange pluses:* the StemCenterNet variant reading the native per-instrument stems. *Red squares:* the clean *Euclid* Q1 MER catalog (VIS-detected, non-spurious). The white bar is 15″.

coder itself is never updated, and the heads read its precomputed feature maps, so head training never runs the foundation at all.

We also tried a Detection Transformer (DETR; ?), which represents objects with a fixed set of learned "query" slots assigned to sources through a matching step. Its set-prediction machinery is built for the complexity of natural scenes and is data-hungry, and on our limited training set it converged slowly and unstably. A per-pixel design fits the problem better: every pixel receives a direct training signal, and detection reduces to finding peaks in a learned source-evidence image, much as sources are found classically. Blending, the main concern for such peak-based approaches, is also less severe here, since the fused representation brings the sharp *Euclid* VIS scaffold to bear alongside the ground-based bands: the deconfusion of seeing-limited imaging by space-based resolution that joint processing was conceived to deliver (?), supplied here by the representation itself rather than by explicit positional priors, joint scene models (e.g., SCARLET; ?), or a dedicated deblending network (?).

A natural source of labels would be the *Euclid* Q1 MER catalog, but we deliberately did not train on it. That catalog is the product of substantial detection machinery: SOURCEEXTRACTOR++ run in a simulation-calibrated dual-threshold configuration, separately on the VIS mosaic and on a stack of the three NISP bands, followed by a dedicated density-based deblending stage and by spurious-source and bright-star cleaning (?). Matching that reference with far less is part of what this section sets out to show: detection from the frozen features should not require heavy curated supervision, so we keep the labels lightweight and internal, generating them directly from the VIS imaging with standard source extraction (SEP; ?) at 3σ, followed by light cleaning: detections sitting on diffraction-spike ridges are removed, each saturated star is consolidated to a single centroid, and near-duplicates are merged. We compared this against a multi-scale peak-finder; SEP produced the cleaner reference, showing markedly higher agreement with the MER catalog at matched purity, so we adopt it as the primary recipe. The head is trained on these labels.

We use the MER catalog instead as an external cross-check, in the form of a clean reference: its objects detected in VIS and not flagged as spurious. Detection in the MER pipeline is VIS-primary, a choice made there because weak lensing requires a sample with a simple selection function, and its supplementary NISP-stack list contributes objects with no VIS counterpart; these lack the VIS magnitudes and S/N that anchor the binned comparisons below, so the clean reference excludes them. Training the head on MER labels rather than our internal ones gives essentially the same detector, its injection depth within $\sim$0.25 mag, with only a marginal change in catalog-match agreement (itself circular, since a MER-trained model is scored against MER). This is consistent with the role the reference plays here: a VIS-anchored detection catalog rather than a complete truth catalog, it neither adds information beyond our VIS-based labels nor rewards the deeper, multi-band sources the head recovers that are faint in VIS but supported by the other bands.

Figure 5 overlays the four source lists on a single held-out *Euclid* VIS tile. The overall agreement is close: on the great majority of sources the four lists coincide. Two differences are visible on close inspection. First, the classical SEP list, even after the cleaning described above, tends to over-fragment extended galaxies, placing several detections across the spiral arms or star-forming clumps of a single resolved system, whereas the learned heads, though trained on those same labels, more often return one detection per object. Second, among the learned heads the fused CenterNet variant is the strongest: it recovers more faint sources than either the classical list or the native-stem variant, including objects that are faint in VIS but supported by the other bands. Whether the detector's added faint sources are real rather than noise is what the injection test below quantifies, by showing that real faint multi-band sources are in fact recoverable; the overlay shows the qualitative pattern, and the quantitative comparison follows.

We attach the detection head to the frozen foundation at the fused, multi-instrument bottleneck (the $0.4''\,\mathrm{px}^{-1}$ latent of Figure 2) and train only the head. Depth is measured by injection, since catalog matching cannot probe past the reference catalog's own incompleteness at the faint end: using source recycling, we take real, isolated, unresolved donors with $19.5 < \mathrm{VIS} < 22.5$, whose magnitudes are taken from MER photometry in a regime where it is secure and whose star-like profiles are enforced by a cut on measured light concentration (the bright catalog is dominated by compact galaxies, whose finite size would otherwise contaminate a point-source measurement), extract their full ten-band cutouts (real PSF and SED), dim them to a fainter target magnitude, inject them at empty positions, selected to have valid coverage and to lie at least $\sim 1''$ from any MER source or pre-injection detection, and re-run the frozen detector, counting an injection as recovered when a new detection appears within $0.3''$. Because recovery is scored against the injected truth, the resulting completeness curve is independent of any catalog's completeness and

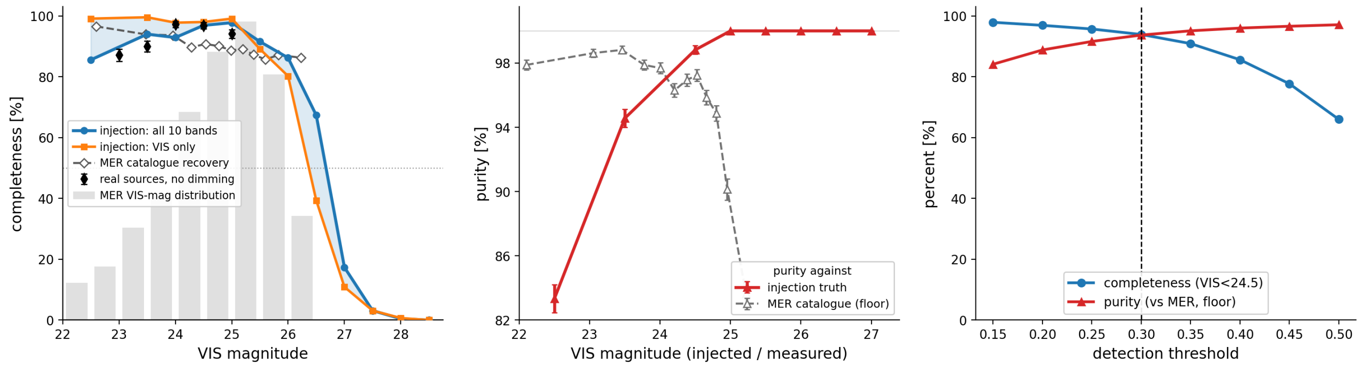

Figure 6. Detection performance of the fused CenterNet head on 108 held-out tiles. *Left:* completeness versus VIS magnitude from source-recycling injection: real unresolved donors, carrying their full ten-band cutouts, dimmed to each target magnitude and placed in all ten bands (blue) or in VIS only (orange); the shaded band is the ≈0.3 mag fusion gain, giving a 50% point-source depth of VIS ≈26.7. Black diamonds are real sources re-injected at native brightness, validating the dimming; their bright-end dip, shared with the all-band curve while in-situ MER recovery holds at ∼97%, is an artifact of the copied stamps, not of the detector on real sky. The dashed curve is in-situ recovery of the clean Q1 MER catalog and the gray histogram its VIS-magnitude distribution; a zero-flux control bounds the protocol's chance floor below 0.66% (95% CL). *Center:* purity against injection truth (filled), 100% fainter than 25 with the bright-end decline from bright-star wing detections, and against the full MER catalog (open), a floor whose faint-end drop tracks the MER turnover. *Right:* completeness (VIS < 24.5) and MER-floor purity versus detection threshold; the dashed line marks the 0.30 working point.

can probe past it; only the magnitude scale is inherited, through the donors, from MER's bright-end photometric calibration. The dimming itself is validated directly: real sources re-injected at their native brightness, with no scaling applied, reproduce the dimmed-donor completeness at 23–25, where the reference catalog is still complete enough to supply them (Figure 6, top). The faint end of the curve carries a calibrated zero: extending the grid past VIS 28 leaves at most 3 recoveries of 450 per magnitude, zero beyond 28.5, and a zero-flux control, donors dimmed by a factor of 10^{-6} and processed identically, is never recovered (0/450 in every mode), bounding the chance-coincidence floor of the protocol below 0.66% (95% confidence); the few recoveries at 28 sit within that floor. Recoveries below the nominal depth are therefore genuine detections of the injected light, and the completeness curve can be read at any S/N, however far below the single-band limits, without a catalog to vouch for it. The fused head reaches a 50% point-source depth of VIS ≈ 26.7, about 0.3 mag deeper than the same detector restricted to VIS alone (26.4), a multi-band gain that carries the census well past the turnover of the MER reference. Repeating the measurement with the compact-galaxy donors instead gives the completeness for moderately extended profiles dimmed to each magnitude, 25.2 against 24.7: shallower in absolute terms, as expected once the light is spread over a larger area, but with a comparable +0.5 mag fusion gain, so the multi-band advantage is not a property of one morphology. Running the identical benchmark on the native-stem head (StemCenterNet) confirms that this gain belongs to the fused representation, not to the head: it reaches a comparable VIS-driven depth but gains nothing from the other nine bands (+0.0 mag vs +0.3 mag; Table 1). We therefore adopt the fused CenterNet configuration for all subsequent results.

To localize where this sensitivity comes from, we repeat the injection placing the source in only one instrument subset at a time; the detector itself always runs on the full ten-band input, so these modes control which bands carry the injected source, not what the head sees. Injecting into the NISP bands alone yields only marginal recovery, peaking below ∼10%; injecting into the Rubin bands alone yields essentially no detections. The head is *Euclid*/VIS-anchored, operating at VIS resolution and trained on VIS-detected labels, so the other bands add depth in fusion but cannot drive detections on their own.

Purity is measured the same way as depth, by injection, because inside an injected scene the truth is complete: every detection that appears after an injection pass can be classified with no external catalog. With one magnitude injected per detector pass (1620 injections per magnitude on the held-out tiles, isolated star donors in 6″ apodized stamps), each new detection is either the injected source itself, a pre-existing sub-threshold peak pushed over the working point by the changed scene (identified with a pre-pass at a low score floor and counted separately), or an induced false positive. Measured this way (Figure 6, center), the detector adds essentially no false positives of its own: purity against injection truth is 100% at every magnitude fainter than 25, with no artifacts among the 1620 injections per magnitude, and the only appreciable impurity is astrophysical, secondary detections on the wings and halos of bright stars (83% at VIS 22.5, contributed mostly by the NISP and Rubin bands), the same population that classical pipelines mask and flag. The zero-flux control anchors the blank-sky limit: donors dimmed by a factor of 10^{-6} yield no detections at 450 empty, coverage-checked positions per mode, so noise peaks on empty sky contribute at neither end.

The MER catalog supplies the complementary in-situ cross-check, real sources at their true positions with whatever blending and neighbors accompany them. Against the clean reference the detector recovers ∼85–97% of listed sources across the magnitude range MER reaches (Figure 6, left, dashed curve); re-binned by measured aperture S/N instead of magnitude, so that no photometric zeropoint enters, the recovery behaves identically. Purity measured against the full MER catalog rises with the detection threshold as completeness falls (Figure 6, right); we adopt a working point of 0.30. Because the MER reference is a detection catalog with its own VIS-primary selection rather than a complete truth catalog, these numbers describe agreement with that reference: the unmatched detections include real multi-band sources absent from MER, carrying Rubin/NISP aperture flux well above random sky positions, so the catalog-based purity is a floor beneath the injection measurement above.

In summary, the head begins from classical VIS detection and extends it: it is deeper, uses all ten bands, and returns one detection per object where classical source extraction fragments extended systems. Its main limitation lies at the other end of the size distribution: large, low-concentration galaxies whose light is spread over many pixels are occasionally missed, because the $0.4''\,\mathrm{px}^{-1}$ fused bottleneck on which the head operates cannot concentrate their flux into a sharp peak. Since simply retraining the encoder at a finer fused scale leaves the representation unchanged (Section 3), the natural next step is a multi-scale readout for the head itself.

5. ASTROMETRY

Astrometry is the second measurement we draw from the frozen foundation, and the question it asks is whether the joint learned representation can improve astrometry itself: both the precision of individual source positions and the alignment between the two surveys. Rubin and *Euclid* each arrive with a mature astrometric solution, independently registered to the Gaia reference frame (**??**), and this work does not attempt to improve either. But nothing in what follows relies on that prior registration. The per-source correction and the cross-survey alignment developed here are built entirely from the pixels: the representation learns where the two instruments see the same sources and uses that correspondence directly. In principle the same machinery would align two surveys that share no external reference frame at all; the Gaia registration matters only for tying the result to an absolute frame, which we do not address here. What the shared frame does provide in this case is a stringent starting point, since two independently Gaia-aligned surveys should already agree, and measuring how well they actually do is itself informative. Concretely, this section develops two products: a per-source position head that transfers the sharp VIS localization to the nine other bands, where the PSF is broader or the source fainter, and a measurement of the residual agreement between the two solutions, which we call the cross-survey concordance field. The name is inherited from the concordance frame of **?**: cross-survey astrometric agreement at the ∼10 mas level, matching each survey's internal precision, was the threshold that study set for joint measurement to proceed, and the field mapped below is a first measurement of where two independently calibrated solutions actually stand against it. The two products live at very different amplitudes, and separating them cleanly is the main work of this section.

Figure 7 shows the starting point: the raw astrometric agreement between the surveys before any correction, from classical Gaussian-fit centroids of sources matched between each band and VIS, pooled over the training field (0.4–2.7×10^4 matched pairs per band, $\sim 1.5 \times 10^5$ in total). Bright sources (S/N ≥ 30; $\sim 10^2$–2×10^3 per band) already agree well, with median per-source offsets of ∼10–25 mas per band, but their offset clouds do not close on the origin: the center of each cloud is displaced from zero by ∼7–10 mas, a coherent systematic distinct from the per-source scatter, most of it a common shift shared across all bands (the concordance field measured below). The NISP cloud centers sit slightly farther from the origin (∼10 mas) than the Rubin ones (∼7 mas) despite the NISP mosaics being delivered on the same pixel grid as VIS; a plausible origin for the small excess is the extra processing the NISP imaging undergoes before reaching that grid, the registration and resampling of the native $0.3''$ frames onto the $0.1''$ MER mosaics (Section 2), though the mosaics alone do not let us isolate the cause. The full source population, dominated by faint galaxies, behaves differently: its median per-source offset is ∼45–55 mas, and the right panel shows that this disagreement follows the expected photon-noise scaling of centroid precision with FWHM and S/N (**?**), drawn there as the dashed guide. The guide's normalization is indicative rather than exact: it adopts a single $1''$

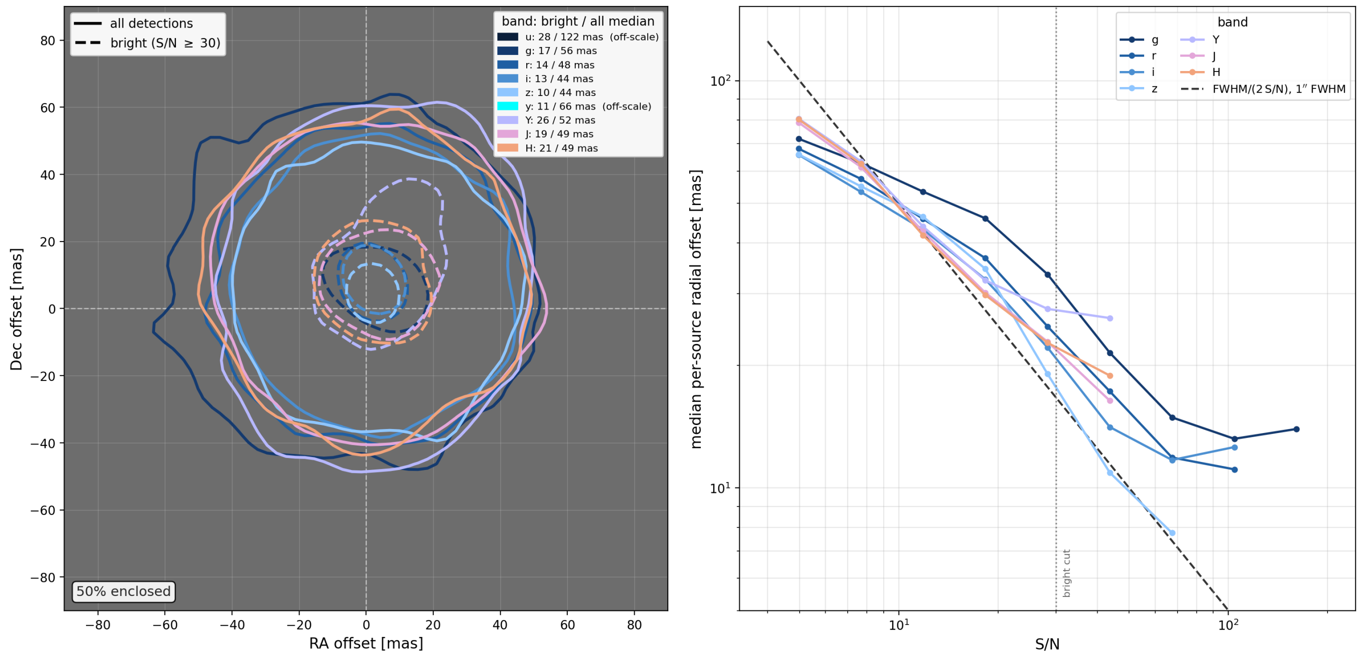

Figure 7. Raw astrometric agreement between Rubin/NISP and *Euclid* VIS before any correction, from classical Gaussian-centroid positions of matched sources pooled over the ECDFS field; plotted offsets are (Rubin/NISP − VIS). *Left:* two-dimensional offset clouds, each contour enclosing 50% of that band's sources; solid contours are all detections, dashed contours bright sources (S/N $\geq$ 30). Although both surveys are independently registered to *Gaia*, the bright clouds do not close on the origin: their centers are displaced by $\sim$7–10 mas, most of it a common shift shared across all ten bands, coherent but not constant over the footprint (the concordance field mapped in Figure 10). The NISP cloud centers sit slightly farther ($\sim$10 mas) than the Rubin ones ($\sim$7 mas; see text). The legend lists the bright/all median per-source offset per band; the u and y clouds lie off the plotted scale and are given as numbers only. *Right:* median per-source radial offset versus S/N for the seven well-measured bands; the dashed line is the photon-noise scaling of centroid precision, FWHM/(2 S/N) for a 1″ FWHM (**?**), drawn as a guide (see text). The dotted line marks the S/N = 30 bright cut.

FWHM for all bands, the aperture S/N on the horizontal axis understates the total-photon S/N of the underlying derivation, and the faintest bins are trimmed by selection, since a source must be detected and matched to enter the sample at all. These conventions let the Rubin curves sit slightly below the line without any centroid outperforming the physical limit, while the NISP curves sit above the lower expectation their sharper PSF would set, plausibly reflecting the undersampled native 0.3″ imaging and the correlated noise introduced when it is resampled onto the MER grid. What the comparison establishes is the slope: all bands follow the $\sim$1/(S/N) decline through the faint regime, identifying the disagreement of the full population as photon noise rather than a calibration failure, before flattening onto the bright-source floor, set by the cloud-center displacement and the VIS centroid error, which photons cannot reduce. The raw offset is therefore not one effect but two, different in nature and very different in amplitude.

This decomposition dictates the strategy. The coherent component is small, a few mas, and is carried by bright sources that are sparse on the sky ($\sim$1–3 arcmin^{-2}); it is a smooth field to be mapped. The dominant component is per-source centering noise: it lives at the photon-noise floor set by each band's PSF and depth, it is uncorrelated between neighboring sources, and no global correction can remove it. It has to be corrected one source at a time, conditioned on the actual image content, which is precisely what a head on the foundation representation can do.

The position head is a small network ($\sim$0.7M parameters) attached to the frozen encoder. For each detected source it reads two local windows: one from the fused bottleneck, whose 0.4″ px^{-1} features carry the chromatic morphology of all ten bands, and one from the VIS stem features at their native 0.1″ px^{-1} sampling, which provide the fine spatial reference. From these it predicts a per-source, per-band position offset together with its uncertainty. The prediction is anchored to the centroid convention itself: in its forward pass the head computes an iterated Gaussian-weighted centroid on the raw VIS window at the query, along with that centroid's photon-noise uncertainty from the same pixels, and the network contributes only a residual bounded at a few times the

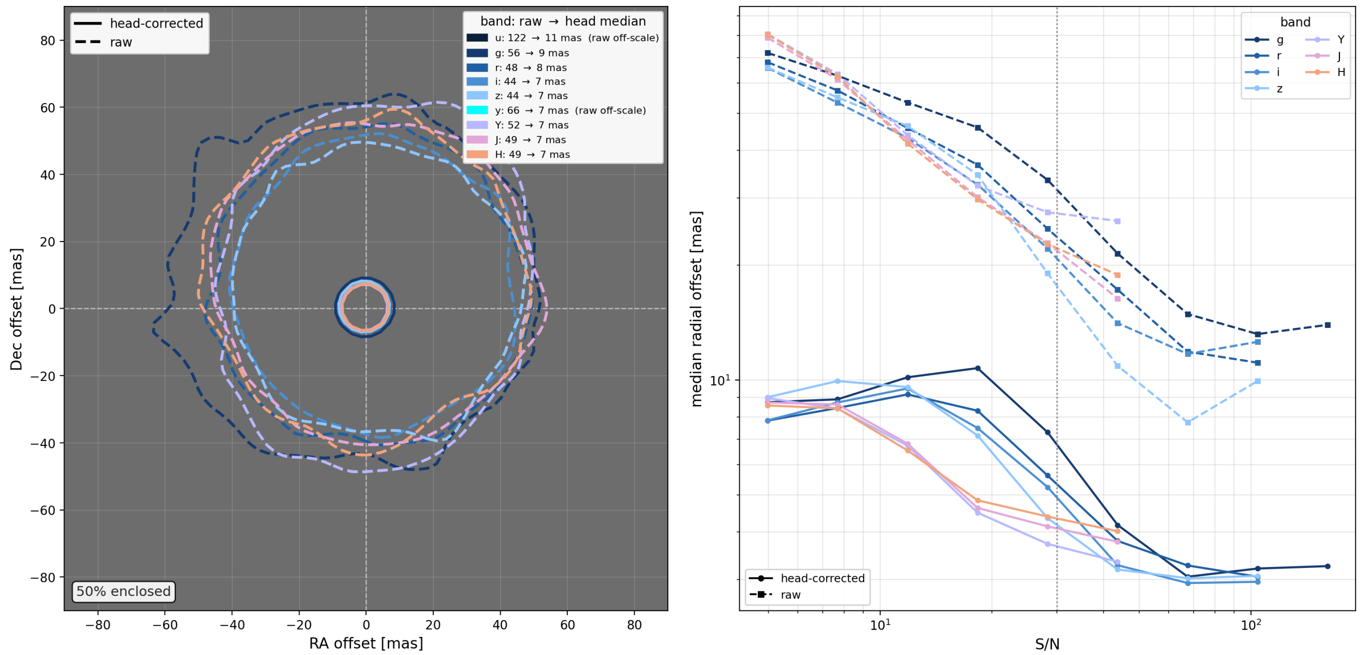

Figure 8. Effect of the per-source position head on cross-survey agreement, using the same offset measurement as Figure 7 (Rubin/NISP − *Euclid* VIS) over all 790 ECDFS tiles. Solid curves are head-corrected positions; dashed curves are the raw classical centroids. *Left:* all-detection offset clouds, each contour enclosing 50% of that band's sources. The head collapses the raw ∼50 mas clouds to ∼7–11 mas and re-centers them on the origin, removing both the per-source scatter and the systematic offset; the legend gives the raw → head median per band (u and y raw clouds lie off the plotted scale). *Right:* median per-source radial offset versus S/N: the head-corrected curves (solid) are far flatter than the raw (dashed), staying near the bright-source floor across the full range, with the in-model centroid anchor holding the bright end at the classical systematics floor. These medians measure agreement with the VIS-centroid convention that also supplies the head's training labels, so they can sit below the accuracy against truth, which is bounded by the VIS anchor and measured directly in Figure 9.

anchor noise. On bright compact sources, where the anchor is precise to a few mas, the network can only polish it; on faint sources, where the anchor is photon-starved, the bound opens and the learned correction carries the measurement. Training is self-supervised and requires no external catalog: VIS Gaussian-fit centroids serve as targets, the query position is perturbed by a controlled jitter, and the head learns to recover the centroid that the local features encode. The loss is a Rayleigh negative log-likelihood that includes a fixed label-noise floor of 5 mas, reflecting the systematic floor of the underlying WCS solutions, so that the head is not rewarded for chasing noise in individual bright-source labels. The predicted uncertainty serves to weight this likelihood, letting the head discount ambiguous sources during training; its calibration as a delivered error bar is checked against the observed residuals below.

Figure 8 shows the effect. The head collapses the raw ∼50 mas offset clouds to ∼7–11 mas medians in every non-VIS band and re-centers them on the origin, removing the per-source scatter and the coherent band offsets together. The S/N dependence flattens correspondingly: faint sources are brought down toward the bright-source floor, largely removing the FWHM/S/N degradation, though a residual dependence remains. Because the head is trained against classical VIS centroids, agreement with those labels could in principle reflect emulation of the label convention rather than genuinely better positions; we test this directly by injecting sources of known position into all ten bands of held-out tiles (Figure 9). At a peak S/N of 5, head positions err against truth by ∼19 mas, compared to ∼50 mas for classical single-band centroids, close to the ∼17 mas floor set by the VIS anchor itself. The mechanism is exactly the transfer the section set out to test: the head carries VIS's localization to the other bands, and it adds no precision beyond what VIS itself provides.

Agreement with the training labels could still conceal a failure of the estimator itself, and an earlier, unanchored version of this head exhibited exactly one: reading the same features and trained on the same labels, it learned its own morphology-weighted center convention and displaced a small coherent population, bright stars and structured galaxies, by tens of mas, repeatedly across bands and across independently trained models. The anchor removes the freedom that failure exploited,

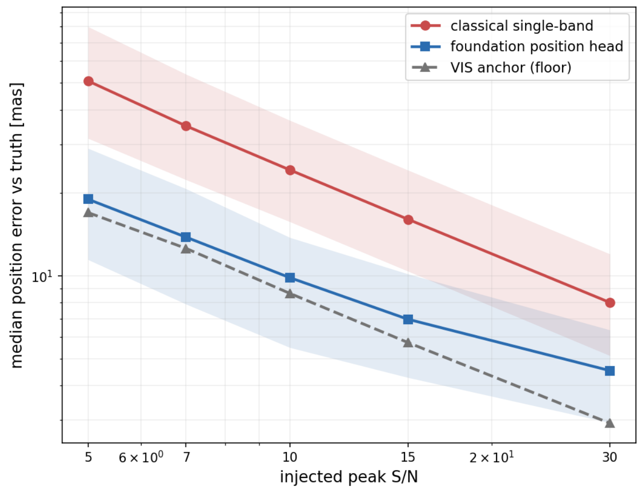

Figure 9. Injection-vs-truth astrometry: the position head transfers VIS-level localization to the Rubin bands. Sources of known position are injected into all ten bands of held-out tiles at controlled peak S/N; the median radial error is measured against the *injected truth* (not the VIS label) and pooled over the Rubin optical bands (*griz*), with shaded 25–75% ranges. The classical single-band Gaussian centroid (red) rises toward faint sources as expected for photon-noise-limited centroids, reaching $\sim$50 mas at S/N = 5. The foundation position head (blue) instead tracks the VIS anchor error (gray, dashed), the floor set by the VIS centroid the head is trained to reproduce, collapsing the S/N = 5 error to $\sim$19 mas, within $\sim$2 mas of the $\sim$17 mas VIS floor.

since the forward-pass centroid pins every prediction to the label convention within the anchor's own noise; the coherent population shrinks by an order of magnitude, and what remains is confined to faint sources where the bound is open. Gaia DR3, propagated by proper motion to the effective epoch of the coadds, supplies a label-independent sanity check on bright stars: within the precision of this small subset, the anchored head remains at the same Gaia-tied floor as the classical centroids, so the amortized estimator shows no bright-star degradation while retaining its factor of several gain for the faint majority. The predicted uncertainty is conservative across S/N after a single empirical rescaling, which we use when interpreting it as the per-source error bar on delivered positions.

With the per-source scatter under control, the small coherent component can be measured cleanly. Figure 10 shows the concordance field, fit to the S/N-weighted offsets with two solvers of deliberately different character: a physics-informed neural network (PINN; **?**) whose auxiliary losses impose curl-free structure, smoothness, and band consistency, and a hierarchical Gaussian process (HGP) that decomposes the field into a component common to all bands, per-instrument group terms, and per-band residuals, with posterior uncertainties that are calibrated empirically on held-out anchors; the HGP expands each of these components on a two-scale spatial basis with correlation lengths of 5′ and 15′, under prior amplitudes of 4, 2, and 1 mas for the common, instrument, and band components, while the PINN's spatial resolution is set implicitly by its smoothness penalty rather than by an explicit basis. The point of using two solvers is that they can fail differently: the PINN constrains the field through physical priors, the HGP through hierarchical smoothness and posterior uncertainty, and structure that survives both is unlikely to be an artifact of either. On the coarse scales the footprint probes, from the 5′–15′ scales of the HGP basis up to the few-tenths-of-a-degree extent of the field, they recover the same $\sim$9–10 mas pattern. Below those scales the comparison is no longer meaningful: refitting the PINN on identical inputs reproduces its own field only to $\sim$5 mas, so finer differences between the solvers are within the fit noise, and we do not interpret structure there. The 5′ basis scale is itself matched to the information the anchors carry: at the $\sim$50–60 deduplicated sources per square arcminute of this footprint (all bands combined), each with tens of mas of centering noise, the mean offset in a single square arcminute is uncertain at roughly the field's own amplitude, and only patches several arcminutes across average it down to the 1–2 mas level where the field stands clear, so no solver could resolve finer structure from these data.

The HGP decomposition shows the field is dominated by its common component ($\sim$7 mas), shared across all ten bands: most of it is a single cross-survey systematic between the two Gaia-tied solutions rather than per-band noise. Stated chromatically, the field is nearly achromatic: for a typical Rubin band the common term carries roughly 80% of the field variance, pointing to a geometric origin, a registration offset between the delivered frames, rather than a wavelength-dependent one. This also bounds atmospheric chromatic systematics: any residual of differential chromatic refraction surviving in the Rubin coadd astrometry must live in the per-band terms, which sit at only 1.5–3 mas across *ugrizy*, while the larger NISP per-band residuals track the resampling excess already noted for Figure 7. The field is stable to the source selection: refitting on only sources with S/N > 10 leaves both its amplitude and its morphology essentially unchanged (per-component amplitudes within $\sim$0.5 mas and spatial correlation >0.96 with the fiducial fit), confirming that it is a coherent systematic rather than an artifact of the faint, high-scatter majority. After the per-source correction the coherent field collapses to $\sim$1 mas, absorbed source by source; in

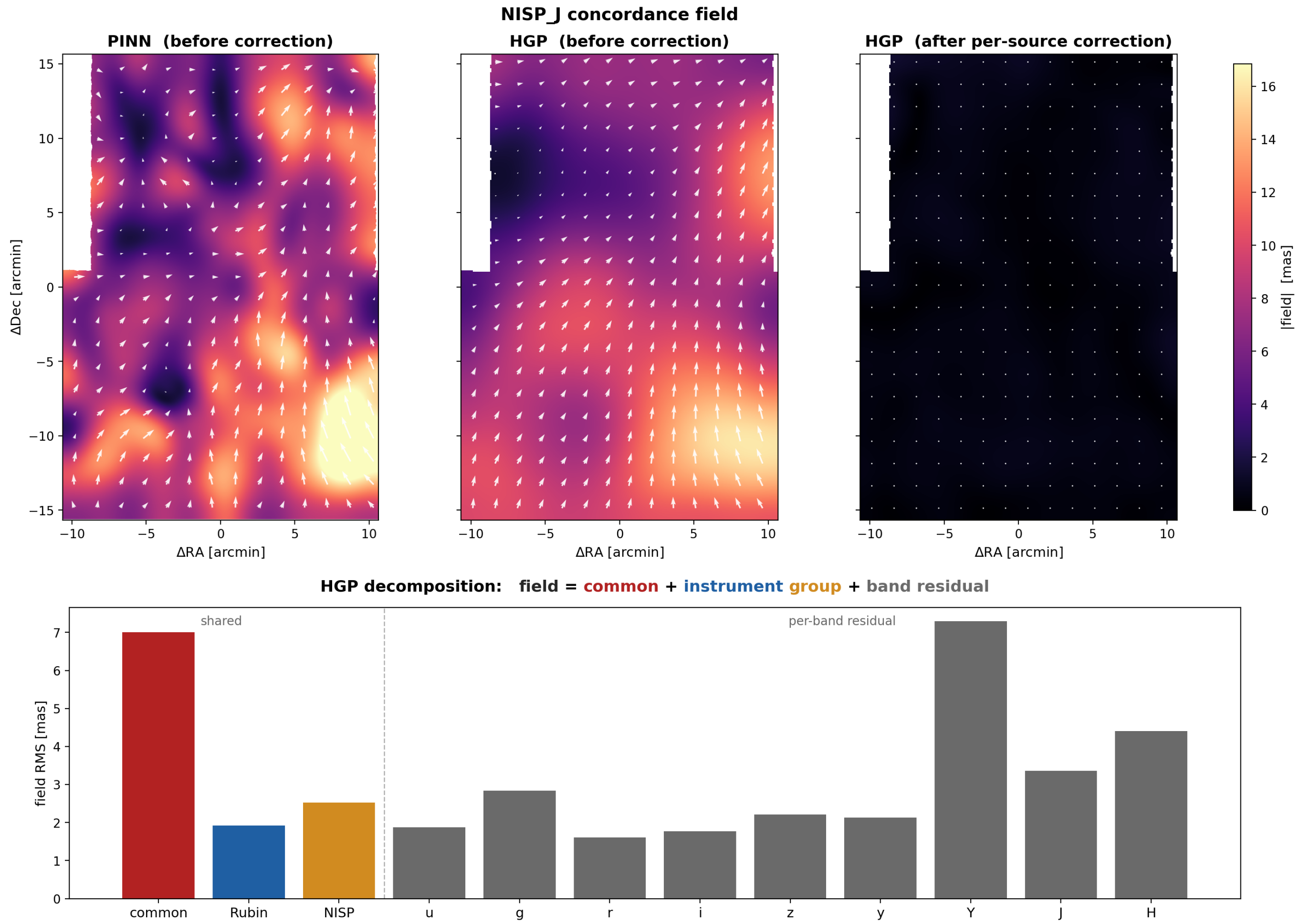

Figure 10. The cross-survey concordance field before and after the per-source correction. *Top:* the NISP J residual field over the ECDFS footprint (arrows: direction; color: |field|; smoothed to $60''$). Before any correction, the field is measured with two solvers: the physics-informed network (PINN, left) and the hierarchical Gaussian process (HGP, center). They recover the same $\sim$9–10 mas pattern, the local amplitude drifting smoothly between $\sim$5 and $\sim$13 mas across the footprint. After the per-source correction (right, same color scale) the coherent field collapses to $\sim$1 mas, absorbed source by source. *Bottom:* the HGP hierarchical decomposition of the pre-correction field into a component common to all bands ($\sim$7 mas), per-instrument group terms (Rubin, NISP), and small per-band residuals (u–H); the dominant common term shows most of the field is a single shared cross-survey systematic rather than per-band noise.

this low-amplitude residual regime the HGP posterior scale requires an empirical factor-of-two calibration on held-out anchors, so we interpret the residual field amplitude more robustly than the raw posterior error bars. The fitted field also closes the loop on Figure 7: subtracting it at the bright-source positions re-centers their offset clouds from $\sim$8 to $\sim$1 mas, with the common component alone accounting for nearly all of the shift; it does so even at held-out positions, provided the held-out gaps are small compared with the HGP's $5'$–$15'$ correlation scales. The bright-source offset and the mapped field are thus the same systematic measured two ways, and since the fit is dominated by the faint majority, bright and faint sources evidently share it. The concordance field is nonetheless a measurement more than a correction, in two specific senses. For the faint majority, the $\sim$50 mas per-source scatter buries the few-mas coherent term, so subtracting the field leaves the pooled median offset unchanged. And the field is coherent but not constant: its local value drifts across the footprint, so it does not extrapolate: refit with an entire sky patch excluded, it fails to re-center that patch's bright clouds. Its value is as a map of how well two independently calibrated surveys agree on the sky, at the few-mas level on the coarse scales this footprint probes.

6. OUT-OF-DISTRIBUTION TRANSFER

The final question is whether any of this transfers. Everything so far (the reconstruction, the detection head, the astrometry head, the concordance field) was mea-

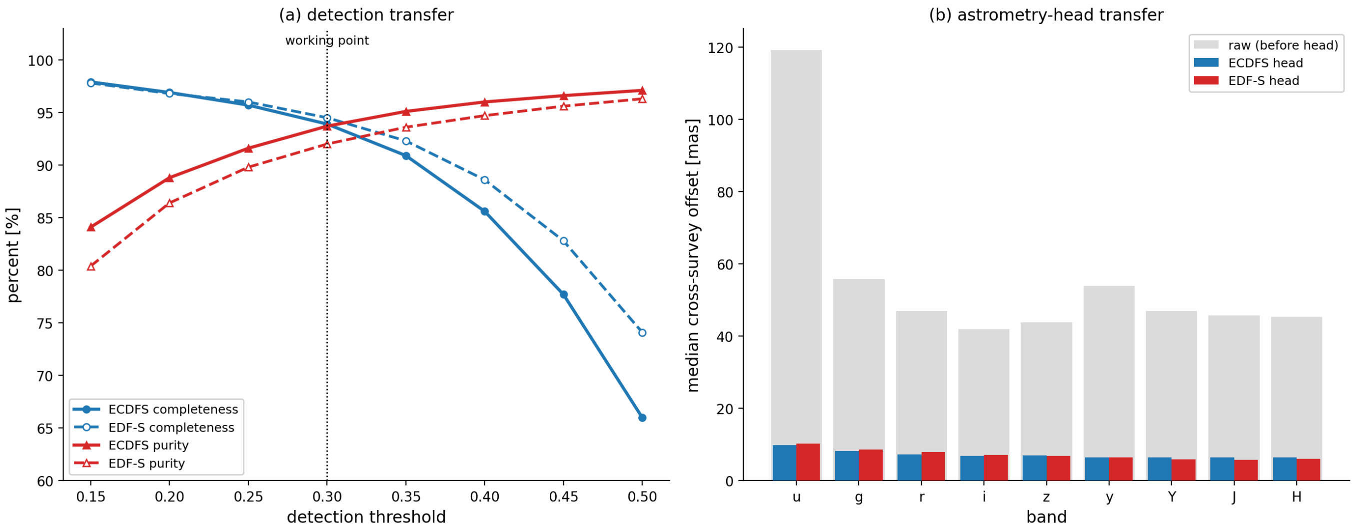

Figure 11. Out-of-distribution transfer to EDF-S with no retraining. The frozen foundation and the detection and astrometry heads, trained only on the ECDFS field, are applied to the held-out EDF-S field (72 tiles, a separate *Euclid* Deep Field never seen in training). *(a) Detection:* completeness and purity versus detection threshold, in-distribution (ECDFS, solid) and out-of-distribution (EDF-S, dashed), scored against each field's clean *Euclid* Q1 MER catalog. Both transfer almost exactly: at the 0.30 working point EDF-S reaches completeness/purity of 94.5%/92.0% against ECDFS 93.9%/93.7%. *(b) Astrometry:* per-band head-corrected median cross-survey offset, ECDFS versus EDF-S; the gray bars mark the raw pre-head offsets. The position head collapses the raw ∼45–60 mas offsets to the same ∼6–10 mas floor on the unseen field.

sured on ECDFS, the field the foundation was pretrained on. A representation that only works where it was trained is of limited use; the point of a foundation model is that the frozen features carry general structure. We test this directly on EDF-S, the Euclid Deep Field South: a widely separated region of sky, processed identically but never seen during pretraining or head training. Nothing is changed: the same frozen foundation, the same detection and astrometry heads exactly as trained on ECDFS, applied to the 72 EDF-S tiles with no retraining or fine-tuning (Figure 11).

Detection transfers almost exactly. Running the production CenterNet head at the same 0.30 working point, completeness against the clean EDF-S MER catalog is 94.5%, matching the 93.9% measured in-distribution, and the completeness-versus-threshold curve tracks the ECDFS one across the full range (Figure 11a). Purity is 92.0%, measured within the MER catalog footprint, again close to the in-distribution 93.7%. The truth-based depth transfers as well: repeating the star-donor injection protocol of Section 4 on the EDF-S tiles covered by its MER catalog yields a 50% point-source depth of VIS ≈26.8 (26.6 from VIS alone), fully consistent with the in-distribution 26.7 (26.4). Detection therefore carries over to a new field with no measurable loss, whether scored against the reference catalog or against injected truth.

The astrometry head transfers just as cleanly. Applied to EDF-S, the raw cross-survey offsets are ∼45–60 mas in $g/r/i/z/y$ and the NISP bands, comparable to ECDFS, while the much shallower u band starts higher; the head collapses them to per-band medians of ∼7–9 mas ($g/r/i/z$) and ∼6 mas (NISP $Y/J/H$), with u highest at ∼10 mas, all within ∼1 mas of the in-distribution floor (Figure 11b). The transfer also holds against truth rather than labels: sources of known position injected into the EDF-S tiles are recovered to ∼17 mas at S/N = 5, within ∼2 mas of the local VIS anchor floor (15.4 mas), reproducing the in-distribution injection result on sky the head has never seen. The alignment learned on one field applies unchanged to another, which is expected if the head is reading a genuine, instrument-level relationship between the bands rather than a field-specific pattern. The anchor contributes to that robustness by construction: it is computed from the pixels of each new field rather than learned from the training one, so the convention the head is tied to travels with the data.

Two points are worth making about what this does and does not show. It is a same-instrument, same-processing transfer test: EDF-S is a different patch of sky, not different hardware or a different pipeline, so it probes generalization across the sky rather than across surveys. And it uses only currently public Q1 data over a single Rubin tract in each of two deep fields, a small fraction of the available Rubin–*Euclid* overlap. That the frozen stack transfers across that gap with no retraining is what makes scaling the same approach to larger areas, and to the deeper releases to come, a matter of applying it rather than rebuilding it.

7. SUMMARY AND DISCUSSION

This work set out to test whether the joint pixel content of Rubin and *Euclid* can be distilled, without supervision, into one representation precise enough for measurement. The foundation model, trained solely to predict each band of the ECDFS imaging from the rest, clears the three checks posed in Section 3.3: held-out bands from both instruments are reproduced with high fidelity, per-band brightness reads out linearly from the frozen features, and reconstruction draws on other bands in proportions that track real spectral relationships. On that footing, the two heads deliver their measurements. The detection head reaches the classical VIS baseline and improves on it in two respects: it reduces source confusion, with each resolved galaxy entering the catalog once, and the full ten-band stack deepens the census by $\approx$0.3 mag over VIS alone, to a 50% point-source depth of VIS $\approx$26.7 (25.2 for moderately extended profiles), an increment the native-stem control traces to the fused features rather than the head design. The astrometry head equips every band with VIS-grade positions: the $\sim$50 mas raw disagreement between matched sources shrinks to $\sim$7–11 mas, and the error against injected truth at S/N = 5 is $\sim$19 mas, nearly saturating the $\sim$17 mas precision of the VIS anchor. For bright stars, whose classical positions already sit at the Gaia-tied floor, the centroid anchor built into the head's forward pass keeps the learned estimator on that floor in the Gaia check, so a single estimator delivers the better of both regimes. Separating off the per-source noise exposes the concordance field: a smooth $\sim$9–10 mas disagreement between two astrometric calibrations that are each tied to Gaia yet offset from one another, the bulk of it a single $\sim$7 mas term present in every band. The whole stack then repeats these results on a second deep field it has never seen, with nothing retrained.

All of this is measured on the earliest public data from both observatories, which sets a floor rather than a ceiling. Rubin DP1 is commissioning imaging with a median PSF of roughly $1''$; operational LSST will deliver sharper images and, over the survey, far deeper stacks. Both improvements propagate directly through the pipeline: a sharper PSF tightens the classical centroids that serve as training labels and as raw baselines alike, and deeper optical imaging may give the Rubin bands enough weight to seed detections by themselves, which at DP1 depth they cannot. *Euclid* Q1 will likewise be superseded by the deeper and better-processed DR1. Near-infrared mosaics at the native NISP sampling are in fact already part of Q1, released as science images without the per-pixel variance maps our inputs require; the deeper releases are expected to provide them fully supported. If the $\sim$3 mas NISP excess of Section 5 does enter with the resampling, it is exactly the kind of inherited preprocessing residual a joint model could then remove, by taking the native-sampling mosaics as input and solving the registration within the model. The same reasoning extends earlier in the pipeline: everything here operates on coadded mosaics, and pushing the representation toward single exposures, where more of the instrumental and atmospheric signal survives, is where the joint approach has the most room to contribute.

The relation between the learned stack and the classical pipelines it builds on is complementary rather than competitive. Classical processing supplied every label and every benchmark in this paper (the survey WCS solutions, the SEP detection labels, the Gaussian-centroid targets, the MER cross-checks), and classical understanding dictated the evaluations that made the learned components trustworthy. When joint processing of these surveys was scoped as a classical pipeline effort, the estimate ran to roughly two hundred work-years of development and upward of a billion CPU hours of computing (**?**); that scope covered the full sky overlap of three surveys and a community science platform, far more than the two heads and single tract demonstrated here, so the numbers do not compare one to one. What the learned stack changes is the cost structure. Pretraining took $\sim$8 GPU-hours, the features are extracted once per dataset, and each new capability is a small head trained on the cached features, under a million parameters for both heads presented here; inference is a single forward pass per tile. This division of labor, one expensive representation feeding many cheap heads, is the practical content of the foundation-model paradigm for survey pipelines.

Label noise, rather than representational capacity, limits several of the in-distribution numbers. Detection depth and purity are both established by injection, with the chance floor calibrated by a zero-flux control, because a reference catalog can only vouch for the sources it contains; scored against the VIS-anchored reference instead, purity reads only as a conservative floor. The astrometric floor is likewise set by the VIS Gaussian-centroid labels the head is trained toward: the injection test places the head within $\sim$2 mas of that anchor, so the head has extracted nearly all the precision its labels contain. The direct implication is that better labels convert into better measurements with no architectural change: deeper VIS stacks, ePSF-based centroiding, or

Gaia-anchored positions would each lower the floor the same head can reach.

The limitations identified along the way point in a consistent direction. The detection head is VIS-anchored (sources injected into NISP alone are recovered only marginally, and into the Rubin bands alone almost not at all), and the $0.4'' \, \mathrm{px}^{-1}$ fused bottleneck under-detects the largest, most diffuse galaxies; a multi-scale bottleneck readout and labels not derived solely from VIS would relax both (the matched-token $0.2'' \, \mathrm{px}^{-1}$ control of Section 3 shows that simply moving the single fused bottleneck to a finer grid is not, by itself, a sufficient fix). The transfer test of Section 6 probes a new field, not new hardware: transfer across instruments, with *Roman* (**?**) as the natural third stream alongside Rubin and *Euclid*, is the harder and more valuable test ahead. The nearest-term step, however, is completing the measurement layer: photometry and shape heads, now in development, will work from the same frozen features and treat the per-band morphology and PSF structure the encoder was forced to retain as the measured quantities themselves rather than as aids to localization. Together with the detection and astrometry heads presented here, these would make the joint Rubin–*Euclid* pixels directly usable for the catalog-level measurements that precision cosmology requires: a single representation, learned once from public imaging, serving every measurement built on it.

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Software: Astropy (**?**), NumPy, SciPy, Matplotlib, PyTorch (**?**), scikit-learn, SEP (**?**), pyvo